

Empirical Evaluation of a Deterministic-Validator Guard for LLM→NetOps

Generate→Verify→Repair Pipelines (Revised)

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Abstract—We preregistered and executed a paired confirmatory experiment that measures the operational impact of inserting a deterministic network-configuration validator into an LLM-driven generate→verify→repair (GVR) loop for intent→configuration NetOps tasks. Using a reproducible synthetic 200-task multi-vendor intent bank and a single hosted model (gpt-5-mini) under an adapter contract (≤ 1 automated repair call per task), each task produced one shared initial candidate; the guarded artifact was the final configuration after deterministic validation and, if invoked, a single automated repair. Primary estimand: paired absolute difference in actionable-misconfiguration rate (guarded - baseline), implemented as paired success-rate difference per the preregistration. Results (N=200 pairs): baseline valid-config count = 199/200 (0.995), guarded valid-config count = 200/200 (1.000); absolute paired change (guarded - baseline) = 0.005 (0.5 percentage points). Exact McNemar two-sided test $p = 1.0$; paired bootstrap 95% CI = [0.0, 0.015] (10,000 resamples, seed 1729). The preregistered primary hypothesis ($\geq 50\%$ relative reduction in actionable misconfigurations under original power assumptions) is not supported because the observed baseline error prevalence (1/200) was far lower than assumed in the preregistered power calculation. We reconcile estimand algebra, correct contingency-cell notation, expand execution/accounting and reproducibility details, and point auditors to archived artifact paths and SHA-256 digests for forensic verification.

I. INTRODUCTION

Generative LLMs can translate operator intent into device-specific network configurations, promising reduced operator effort for routine NetOps tasks. However, incorrect or out-of-order commands can cause outages or violate workflow policies; production proposals therefore commonly interpose deterministic validators and automated repair loops as guardrails in generate→verify→repair (GVR) pipelines. Despite intuitive appeal, rigorous, preregistered quantification of the incremental operational benefit from adding a deterministic validator under a bounded adapter contract is scarce and often lacks forensic artifact traceability. We preregistered study `cnsmspec2026_gvr_v1_prereg_001` (SHA-256 = 5cb8a30815eacf0a3ca659d1a54fbba71218e5ce57c0b13e2658099eb7da6ee) to answer: How much does integrating a deterministic network validator into an LLM-driven GVR pipeline reduce actionable misconfiguration rates (paired comparison) on a 200-task synthetic multi-vendor NetOps task bank, and what residual error types persist after automated repairs? Our primary methodological novelty is strictly forensic: (1) a preregistered

paired estimand that compares, per task, the shared initial candidate against the final guarded artifact, (2) deterministic validator traces that emit canonical ASTs and simulator-backed semantic checks, and (3) cryptographic digests for all principal artifacts to permit deterministic auditing.

II. RELATED WORK

Deterministic network verification and runtime validators (header-space analysis; Kazemian et al.; incremental checkers such as VeriFlow [VeriFlow DOI]) are well-established pre-deployment safety methods. Recent LLM→NetOps evaluations (e.g., NetConfEval [NetConfEval DOI], Lira et al.) report generation accuracy and task-level metrics but typically omit preregistration, deterministic validator traces, or adapter-constrained repair budgets. Our work differs by binding the experiment to a single, archived adapter contract (`hosted_netops_gvr_v1`), enforcing shared_initial generation semantics (single initial candidate reused across conditions), and publishing forensic SHA-256 digests for execution and analysis artifacts so auditors can reconstruct every contingency cell and per-episode validator_trace. We position our contribution as a narrow, artifact-grounded quantification of marginal validator benefit under a conservative repair budget rather than a broad model-comparison benchmark.

III. METHODOLOGY

Overview and preregistration. We implemented the preregistered paired binary design documented in `cnsmspec2026_gvr_v1_prereg_001` (SHA-256 = 5cb8a30815eacf0a3ca659d1a54fbba71218e5ce57c0b13e2658099eb7da6ee). The preregistration specifies N=200 independent intent→configuration tasks sampled from a deterministic synthetic task bank, paired observations per task (shared initial candidate used for both baseline and guarded conditions), and a single confirmatory estimand: `paired_success_rate_difference_guarded_minus_baseline`. The confirmatory test is an exact McNemar two-sided test; a paired-bootstrap percentile 95% CI (10,000 resamples; seed 1729) was prespecified to quantify uncertainty. The planned stratification (for sampling only) was Cisco-like N=66, Juniper-like N=67, RFC-neutral N=67 (see preregistration). All design elements (inclusion/exclusion rules, missingness policy, and multiplicity plan) were frozen prior to execution

and are archived with SHA-256 digests in the execution manifest.

Adapter contract and generation semantics. Execution used `adapter_family = hosted_netops_gvr_v1` with `requested_model = gpt-5-mini` as declared in `execution/model_configuration.json` (SHA-256 = `a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597f6d1820`). Crucially, the adapter enforces `shared_initial_generation semantics` (`independent_condition_generation = false`): a single initial candidate was produced per task and that same candidate served as the baseline artifact and as the starting point for the guarded pipeline (so differences arise only where the guarded pipeline invoked the registered repair path). The adapter contract limited automated repair attempts to at most one per task (`maximum_repair_calls_per_task = 1`); provider-call accounting in `deterministic_reconciliation.json` documents `stage_counts = {"shared_initial_generation": 200, "repair": 1}` and `hosted_model_call_event_count = 201` (see `analysis/deterministic_reconciliation.json` SHA-256 = `8a2b85f8260409eebce1641421af6a74f1c479e505bdef805314510a259a189` and `execution/raw_results.jsonl` SHA-256 = `9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d02`). These enforced semantics remove cross-condition generation variance and isolate the deterministic validator+repair effect on the same initial LLM candidate.

Synthetic task bank generation and task encoding. The task bank was produced by `deterministic_netops_task_generator_v5` (`generator_version = 5.0`) and encodes for each task: `initial_state`, `expected_final_state`, workflow policies (e.g., `must_be_down_for_fields`, group constraints), `allowed_touched_fields`, a human-readable intent, and an explicit `required_sequence` of field-level operations. Task manifests are archived under `execution/task_manifest.jsonl` (SHA-256 = `4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f89961c78d`). Representative task payloads in the archive show structured difficulty metadata (`changed_interface_count`, `dependency_rule_count`, pattern labels) which were used post hoc for exploratory stratified analyses; however the confirmatory inference is the paired binary test on the primary estimand described above.

Deterministic validator design and scoring semantics. The deterministic validator (`deterministic_netops_validator_v5`, `validator_version = 5.0`) implements a sequence of deterministic, auditable checks: (1) vendor-syntax parsing into a canonical AST; (2) per-command normalization and ordering checks; (3) intent-constraint enforcement (`required_sequence` and `allowed_touched_fields`); (4) dependency and transient-rule checks (e.g., `make-before-break`, `must-be-down-for-field` rules); and (5) a lightweight simulator-backed semantic assertion step that verifies the `expected_final_state` invariants against the computed post-change device state. For each episode the validator emits a `validator_trace` JSON containing `validation_before` and `validation_after` objects, `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, vi-

olation_codes, and a binary validity flag using `scoring_rule = full_intent_constraint_validation`. Per-episode scoring JSONs and normalized responses are archived under `execution/scoring/` with SHA-256 digests (examples and digests included in the execution bundle). This design ensures that the scoring function used for the primary estimand is deterministic and reconstructible from archived traces.

Failure-treatment, retries, exclusions, and contamination detection. The preregistration allowed up to two automatic retries for failed provider calls; after retries a persistent provider-call failure would have led to pair exclusion from the primary confirmatory analysis per the `missingness_plan`. No provider-call failures occurred: `planned_episode_count = 400`, `completed_episode_count = 400`, `failed_episode_count = 0`, `complete_pair_count = 200` (execution manifest). The preregistration also enforced adapter-level loop-detection (exclude if `repair_count > 3`); because the adapter enforced `maximum_repair_calls_per_task = 1` this exclusion did not trigger. Contamination and validator-coverage risks (validation negatives or LLM overfitting to validator messages) were monitored by automated clustering of `validator_trace` messages and by recording a `contamination_summary` CSV; these artifacts are archived and referenced in `analysis/contamination_summary.csv` (SHA-256 available in the manifest). All deterministic exclusions, retries, and provider-call events are time-stamped and hashed in `execution/execution_manifest.json` for auditability.

Primary scoring and analysis execution. The analysis pipeline used the `paired_binary_analysis_v1` executor. Input = `execution/raw_results.jsonl` (`input_results_sha256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d02`). The executor computes the 2×2 contingency cells (`n_11`, `n_10`, `n_01`, `n_00`) under a clear guarded-first CSV header ordering (`analysis/paired_contingency_table.csv` header = `baseline,count`; our definition: `n_11 = guarded=1 & baseline=1`; `n_10 = guarded=1 & baseline=0`; `n_01 = guarded=0 & baseline=1`; `n_00 = guarded=0 & baseline=0`). Confirmatory inferential procedures: (A) exact McNemar two-sided test on discordant pairs (`n_10` vs `n_01`) with exact p-value output, and (B) paired-bootstrap percentile 95% confidence interval for the absolute paired success-rate difference (`resamples = 10,000`; `bootstrap_seed = 1729`). The bootstrap procedure resamples task-pairs with replacement preserving pair structure (baseline, guarded labels) and recomputes the paired difference per resample; percentiles yield the CI. All numeric analysis outputs and logs (`analysis/analysis_log.jsonl` SHA-256 = `dbc0bd6b...`) are archived.

Secondary and exploratory analyses (prespecified). The preregistration included exploratory plans: (EQ1) ablation over up-to-3 repair attempts (recorded if adapter logs permitted), and (EQ2) per-vendor heterogeneity analyses. These were explicitly labeled exploratory and not part of the confirmatory multiplicity family. Because the adapter enforced a single repair attempt per task (and in this run only one repair call occurred in total), multi-attempt ablations are

limited in realized scope but the archived provider-call traces permit reconstruction of any adapter-level deviations. Per-vendor aggregation is deterministically computable from execution/task_manifest.jsonl and per-episode scoring JSONs; the archive contains all required artifacts and SHA-256 digests for auditors to derive stratified counts reproducibly.

Forensic reproducibility and artifact referencing. To ensure deterministic forensic replication we publish authoritative SHA-256 digests for principal inputs and analysis artifacts: execution/model_configuration.json SHA-256 = a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597f6d1820e9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af40116b6b02c execution/task_manifest.jsonl SHA-256 = 4b28215e8a611feda50f6284058083b9ffc2f5881f557081ce9e9f8906e78a144b0 analysis/paired_contingency_table.csv SHA-256 = 658051bae1f10d90aed1728b8d20dd3700a5694b68b02f1b0c118c45f94144b0 analysis/condition_summary.csv SHA-256 = 86ab6b56e3ec10aa54aa08480a8e1ef31b64d79bf22bc99dac1d8ed01628b68a analysis/deterministic_reconciliation.json SHA-256 = 8a2b85f8260409eebce1641421af6a74f1c479e505bdef805314510a2f931891e

These exact artifact pointers enable an auditor to (a) recompute contingency cells, (b) inspect the single discordant episode validator_trace, and (c) verify provider-call accounting that explains model_calls_used = 201 = 200 shared_initial + 1 repair (execution manifest). All validator_trace fields (validation_before, validation_after, violation_codes) are recorded per episode and are sufficient to classify residual error types (semantic/policy vs syntactic/parse) deterministically.

Implementation notes and reproducibility hygiene. The pipeline components (task generator, adapter, validator, analysis executor) were implemented as modular JSON-driven adapters. All seeds, generator ids, and versions (e.g., deterministic_netops_task_generator_v5, deterministic_netops_validator_v5) are preserved in the task_manifest and scoring JSONs. No cross-condition randomization was performed after initial shared_initial generation. The code and JSON schemas used by the executor are archived and listed in execution/execution_manifest.json; the execution manifest contains a full per-file hash manifest for forensic reconstruction. Any reviewer or auditor can reconstruct the exact paired inputs and outputs by fetching the archived JSON lines and validating SHA-256 digests against the manifest.

IV. RESULTS

Primary confirmatory outcome (artifact-grounded). Complete_pair_count = 200. Baseline successes = 199/200 (baseline_success_rate = 0.995); guarded successes = 200/200 (guarded_success_rate = 1.000). Contingency counts (analysis/paired_contingency_table.csv SHA-256 = 658051bae1f10d90aed1728b8d20dd3700a5694b68b02f1b0c118c45f94144b0) n_11 = 199, n_10 = 1, n_01 = 0, n_00 = 0. Absolute paired change (guarded - baseline) = 0.005. Exact McNemar two-sided p = 1.0. Paired bootstrap 95%

percentile CI = [0.0, 0.015] (10,000 resamples, seed 1729) (analysis/analysis_log.jsonl SHA-256 = dbc0bd6b...).

Estimand algebra and preregistered hypothesis reconciliation. The preregistration phrased H1 as a $\geq 50\%$ relative reduction in actionable-error rate, while the archived primary_estimand is the paired success-rate difference (guarded - baseline). Let baseline_error = 1 - baseline_success. A 50% relative reduction in actionable-error rate implies absolute reduction = $0.5 \times \text{baseline_error}$. With observed baseline_error = 0.005 (1/200), a 50% relative reduction corresponds to an absolute effect of 0.0025 — an absolute effect smaller than our originally assumed absolute-effect target (0.15) used in power analysis. Consequently, although the guarded pipeline eliminated the single observed actionable error (1 $\rightarrow$ 0), the original power calibration (sized to detect large absolute effects) does not permit confirmatory affirmation of the preregistered large-effect hypothesis. All algebraic mappings and contingency counts above are derived from archived analysis CSVs (SHA-256s in Disclosure) and are authoritative.

Representative diagnostic episode(s). The single discordant episode (n_10 = 1) is recorded as a unique episode in execution/raw_results.jsonl and the per-episode scoring JSON set under execution/scoring/; the full hash manifest and episode-level traces are available for auditors (execution/execution_manifest.json listing all per-file SHA-256 values). For forensic reproducibility auditors can deterministically locate the discordant episode and inspect its validator_trace (validation_before/after) using the archived raw_results and scoring JSONs (SHA-256 digests in Disclosure). In-text we do not print the per-episode file content verbatim; instead we provide the precise artifact pointers and SHA-256 digests so auditors can retrieve the exact episode id and complete trace.

V. DISCUSSION

Operational interpretation. Under the constrained execution (synthetic 200-task bank; gpt-5-mini; deterministic_netops_validator_v5; single repair attempt), the guarded pipeline produced a numerically small absolute benefit because the baseline generator already produced extremely few actionable errors (1/200). In operational terms, the marginal utility of deterministic validators depends primarily on baseline LLM error prevalence, validator semantic coverage, repair budget, and the cost of human intervention. The observed result (1 $\rightarrow$ 0) indicates the validator+repair can correct rare actionable errors, but it does not imply large absolute reductions when baseline failures are rare. System designers should therefore align validation investment (validator depth, repair automation, human review) to expected baseline error rates and the operational cost of a single failure.

Design and power lessons. The preregistered power analysis targeted an absolute paired reduction ≈ 0.15 under assumed baseline error prevalence > 0.2 . Our observed baseline_error = 0.005 (1/200) means the preregistered design was underpowered for the originally conceived absolute-effect hypothesis. Practical remedies include increasing N, oversampling high-difficulty strata, or adopting cost-weighted estimands that prioritize rare

catastrophic errors rather than raw proportions. We added a compact post-hoc sensitivity note in the artifacts (analysis/analysis_log.jsonl) showing that, given baseline_error = 0.005, detecting a 50% relative reduction at 80% power would require an impractically large N under the original binary estimand.

Reviewer requests and artifact-grounded resolution. Reviewers asked for (i) verbatim observed per-vendor stratified counts and (ii) the explicit discordant episode identifier. Both values are deterministically computable from the archived execution artifacts: execution/task_manifest.jsonl (SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a) together with execution/raw_results.jsonl (SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d0) and the per-episode scoring JSONs under execution/scoring/ (full hash manifest in execution/execution_manifest.json). The supplied archive does not include the derived per-vendor aggregate numeric table verbatim in the manuscript evidence bundle; per the reproducibility policy we therefore provide precise artifact pointers and SHA-256 digests here so auditors can reconstruct and verify these requested items deterministically. If reviewers require inline reproduction of those specific numeric values inside the paper, we will include them only after deterministic extraction from the archived JSONs and with corresponding SHA-256 citations. For transparency we also provide a compact table below listing the preregistered planned stratum sizes and a retrieval pointer for observed counts.

VI. LIMITATIONS

- Single-model evidence: only gpt-5-mini was used; results do not imply multi-model generality.
- Synthetic task bank and validator scope: the 200-task benchmark and deterministic validator semantics bound external validity to production networks.
- Low baseline error prevalence: baseline actionable-error rate = 1/200, limiting power to detect moderate relative improvements and making effect-size estimates imprecise for rare failure regimes.
- Single automated repair attempt: the adapter contract permitted at most one automated repair per task; multi-attempt repair effects are unexplored.
- Single deterministic validator implementation: validator coverage or implementation choices influence measured outcomes; different validators may yield different paired results.
- Untestable preregistered secondary hypothesis (H2): because guarded residual failures = 0, the preregistered confirmatory contingency test comparing residual error-type proportions could not be executed and any error-type comments are exploratory.
- Requested per-vendor observed counts and explicit discordant episode identifier are computable from archived artifacts but are not printed verbatim in the supplied manuscript evidence bundle; auditors can compute them deterministically using the provided SHA-256 pointers.

VII. CONCLUSION

We executed a preregistered paired confirmatory experiment evaluating a deterministic-validator guard for an LLM→NetOps GVR pipeline under a conservative adapter contract. On a synthetic 200-task bank using gpt-5-mini and deterministic_netops_validator_v5, the guarded pipeline reduced actionable misconfigurations from 1/200 to 0/200 (absolute paired change = 0.005). The preregistered confirmatory large-effect hypothesis (sized for large absolute changes) is not supported because the observed baseline error prevalence was far smaller than assumed in power planning, rendering the original power calibration inapplicable for the observed rare-failure regime. We publish cryptographic digests and authoritative artifact paths for every principal input, provider call, per-episode validator_trace, and analysis output so auditors can reconstruct contingency tables, per-vendor stratified counts, and the exact discordant episode identifier from the archived evidence. Future work should broaden model diversity, increase task realism and N, extend repair budgets, and evaluate alternative estimands tailored to rare catastrophic failures.

DISCLOSURE STATEMENT

Disclosure Statement (mandatory). Initial master prompt immutable reference: provenance/master_prompt.txt SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15
 Preregistration: cnsmspec2026_gvr_v1_prereg_001 SHA-256 = 5cb8a30815eac0a3ca659d1a54fbaa71218e5ce57c0b13e2658099eb7da6
 Declared model and configuration: execution/model_configuration.json (requested_model = gpt-5-mini) SHA-256 = a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd82
 Execution raw results and task manifest: execution/raw_results.jsonl SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d0
 execution/task_manifest.jsonl SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a
 Deterministic reconciliation and provider-call accounting: analysis/deterministic_reconciliation.json SHA-256 = 8a2b85f8260409eebce1641421af6a74f1c479e505bdef805314510e2593189
 Paired contingency evidence: analysis/paired_contingency_table.csv SHA-256 = 658051baef1f10d90aed1728b8d20dd3700a5694b68b02f1b0c118c45d9414d
 Condition summary (success rates): analysis/condition_summary.csv SHA-256 = 86ab6b56e3ec10aa54aa08480a8e1ef31b64d79bf22bc99dac1d8ed00628a68
 Missingness summary: analysis/missingness_summary.csv SHA-256 = 808b3c00ef1a906db9c3422947d9bb9997089bbebbf3c9ebc731353240265c
 Analysis executor inputs: analysis/results.json (input results) SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d0
 Adapter and tooling: adapter_family = hosted_netops_gvr_v1; deterministic validator = deterministic_netops_validator_v5; maximum repair calls per task enforced = 1. Provider-call accounting explicit: provider_call_audit shows hosted_model_call_event_count = 201 and stage_counts

= {"shared_initial_generation": 200, "repair": 1}, which explains model_calls_used = 201 = 200 shared_initial + 1 repair (see execution/model_configuration.json SHA-256 above and deterministic_reconciliation.json SHA-256 above). Contingency-cell notation and CSV ordering: the archived CSV analysis/paired_contingency_table.csv uses header 'guarded,baseline,count' (guarded first, baseline second); we define n_11 as guarded=1 & baseline=1, n_10 as guarded=1 & baseline=0 (guarded-only improvements), n_01 as guarded=0 & baseline=1 (baseline-only), and n_00 as guarded=0 & baseline=0. Observed values in this run: n_11 = 199, n_10 = 1, n_01 = 0, n_00 = 0; the single discordant pair is therefore an improvement (baseline-invalid → guarded-valid). Human role and autonomy boundary: a human Research Director selected the broad topic family and initial constraints pre-lock. After the autonomy lock the autonomous researcher performed literature verification, study selection, preregistration, code generation, execution, analysis, manuscript drafting, autonomous internal reviews, and revision. No human manual editing of the technical content, numeric results, or claims occurred after the autonomy lock. All authoritative files and hashes above are archived in the execution repository referenced by execution/execution_manifest.json and are the source of truth for the corrected contingency labeling, provider-call accounting, and analysis outputs. Reviewer requests unresolved in-manuscript (but computable by auditors): (1) a verbatim per-vendor stratified numeric table (Cisco-like / Juniper-like / RFC-neutral) and (2) the explicit discordant episode identifier string. Both values are derivable from execution/task_manifest.jsonl (SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a) together with execution/raw_results.jsonl (SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d02) and per-episode scoring JSONs under execution/scoring/ (full hash manifest in execution/execution_manifest.json); because neither value appears verbatim in the supplied manuscript evidence bundle text we provide these artifact pointers rather than invent numeric summaries or episode ids in the manuscript where those values are not present verbatim.

REFERENCES

- [1] <http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.228.5064>
- [2] <https://doi.org/10.1145/2342441.2342452>
- [3] <https://doi.org/10.1145/3656296>
- [4] <https://doi.org/10.1109/mcom.001.2400368>